

1.3. Analytical Errors and Validation of Analytical procedures

Errors in pharmaceutical analysis

- The terminology '*error*' invariably refers to the difference in the numerical values between a measured value and the true value.
- ❖ It has become universally accepted in methods of comparison that the percentage composition of a 'standard sample' provided and certified by the NIST, BPCRS or EPCRS must be regarded and treated as absolutely correct, pure and authentic while evaluating a new analytical method
 - Consequently, the differences thus obtained between the standard values and those by the new analytical methods are then treated as '*errors*' in the latest procedure.

- Analytical errors may be broadly categorized into two categories:-

(i) Determinate (systematic) Errors, and

(ii) Indeterminate (random) Errors.

- It is pertinent to mention here that it becomes rather difficult at times to place a particular ‘error’ into one of the above mentioned categories

i) Determinate (systematic) errors

- These are errors that possess a definite value together with a **reasonable assignable cause**.
- In principle these avoidable errors may be measured and accounted conveniently.

Determinate (systematic) errors:

- (a) **Personal Errors** : due to ‘personal equation’ of an analyst and have no bearing whatsoever either on the prescribed procedure or methodology involved.
- (b) **Instrumental Errors** : due to faulty and uncalibrated instruments, such as : pH meters, single pan electric balances, UV spectrophotometers,
- (c) **Reagent Errors** : due to individual reagents
- For instance : impurities inherently present in reagents ; unwanted introduction of ‘foreign substances’ caused by the action of reagents on either porcelain or glass apparatus.

(d) Constant Errors :

- **Example :** Assuming a constant equivalence point error of 0.10 ml is introduced in a series of titrations, hence for a specific titration needing only 10.0 ml of titrant shall represent a relative error of 1% and only 0.2% for a corresponding 50 ml of titrant consumed

Errors in pharmaceutical...

(f) **Errors due to Methodology** : Both improper (incorrect) sampling and incompleteness of a reaction often lead to serious errors.

- A few typical examples invariably encountered in titrimetric and gravimetric analysis are cited below :

S.No.	Titrimetric Analysis	S.No.	Gravimetric Analysis
1.	Failure of reactions to proceed to completion,	1.	Significant solubility of precipitates,
2.	Involvement of either induced or side-reactions,	2.	Co-precipitation and post-precipitation,
3.	Reactions due to substances other than the one being assayed, and	3.	Decomposition,
4.	A noticeable difference occurring between the stoichiometric equivalence point of a reaction and the observed end-point.	4.	Volatilization of weighing forms on ignition, and
		5.	Precipitation of constituents other than the desired ones.

Indeterminate (random) errors:

- **Cannot be pin-pointed** to any specific well-defined reasons.
- Usually manifested due to the **minute variations** which take place inadvertently in several successive measurements performed by the same analyst, using utmost care, under almost identical experimental parameters.
- Mostly random in nature and ultimately give rise to high as well as low results with equal probability.
- **Neither be corrected nor eliminated**, and therefore, form the ‘**ultimate limitation**’ on the specific measurements.

Minimizing systematic errors

- Systematic errors may be reduced substantially and significantly by adopting one of the following procedures rigidly.

(i) Calibration of Instruments, Apparatus and Applying Necessary

Corrections

- Most of the instruments, commonly used in an analytical laboratory,
- Such as : UV-Spectrophotometer, IR-Spectrophotometer, single—pan electric balance, pH-meter and the like must be calibrated duly, before use so as to eliminate any possible errors.

- In the same manner all apparatus (pipettes, burettes, volumetric flasks, thermometers, weights etc.,) must be calibrated duly, and the necessary corrections incorporated to the original measurements.
- In some specific instances where an error just cannot be avoided it may be convenient to enforce an appropriate correction for the effect that it ultimately causes ;
 - For instance : the inherent impurity present in a weighed precipitate can be estimated first and then deducted duly from its weight.

(ii) Blank Determination :

- In order to ascertain the effect of the impurities present in the reagents employed and reaction vessels used ; a blank determination is an absolute necessity.
- It may be accomplished by performing a separate parallel estimation, without using the sample at all, and under identical experimental parameters as employed in the actual analysis of the given sample.

iii) **Using standard substance**

- External standard
- Internal standard
- Standard addition

External standard

- It essentially comprises of performing an altogether separate estimation under almost **identical experimental parameters** with a quantity of a standard substance that consists of similar weight of the component as is present in the unknown sample.
- Thus, the concentration of the component present in the unknown sample:

$$C_{sample} = \frac{R_{sample}}{R_{std}} C_{std}$$

Internal standard

- Employed if there is **variation in experimental parameters**.
 - The effect of such variations can be eliminated by use of internal std.
- **An equal amount of an internal standard, a component that is not present in the sample, is added to both the sample and standard solutions.**
- Quantification is achieved by using ratios of response of the component to the internal std (R_r):

$$C_{sample} = \frac{R_{r_sample}}{R_{r_std}} C_{std}$$

Standard Addition

- This is especially useful when there is a **problem with interference** from the sample **matrix**, since it cancels out these effects.
- Here, a **small known quantity of the component under estimation is added to the sample**, which is subsequently subjected to analysis for the total amount of component present.
 - The actual difference in the quantity of components present in samples with or without the added component ultimately gives the recovery of the quantum added component.
 - The analyte concentration in the original sample:

$$C_{unknown} = \frac{R_{Unspiked}}{R_{spiked} - R_{unspiked}} C_{spiked}$$

Validation of Analytical procedures

- Validation is the **conformation** of a newly developed analytical methods that is going to be used in QC to produce analytical results as designed and expected.
- The object of validation of an analytical procedure is to demonstrate that it is suitable for its intended purpose determined by means of well-documented experimental studies.
- Accuracy and reliability of the analytical results is crucial for ensuring quality, safety and efficacy of pharmaceuticals.
 - For this reason, regulatory requirements have been published for many years.

General recommendation in method validation

- The principle of the test procedure should be described briefly.
- Procedure should be sufficiently detailed to make repetition by experts or other analysts

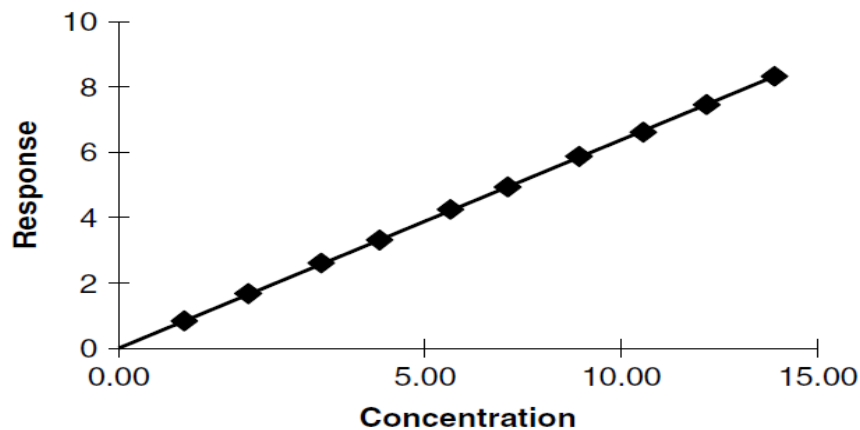
& the following description should be provided:

- ✓ Parameters evaluated or tested
- ✓ Reagents preparation techniques
- ✓ Calculation technique of assessed parameters
- ✓ Equipment and parameters
- ✓ Reference standard used
- ✓ Precaution to be taken

❖ The common parameters that should be verified in method validation:

Linearity

- The ICH defines the *linearity of an analytical procedure as the ability (within a given range) to obtain test results of variable data which are **directly proportional to the concentration (amount of analyte) in the sample.***
- The equation of a straight line takes the form: $y = ax + b$,
 - Where b intercept of y-axis, a slope of the line.
- At least five concentration levels should be used.
- Under normal circumstances, linearity is achieved when the coefficient of determination (r^2) is ≥ 0.997



Accuracy

- The ICH defines the *accuracy of an analytical procedure as the **closeness of agreement between the values*** that are accepted either as conventional true values or an accepted reference value and the value found.
- Accuracy is usually reported as **percent recovery** by assay, using the proposed analytical procedure, of known amount of analyte added to the sample.
- Typical accuracy of the recovery of the drug substance in the mixture is expected to be about 98 to 102%.
- Values of accuracy of the recovery data beyond this range need to be investigated.

Precision

- Precision is the degree of agreement among a series of measurements of the same quantity; it is a measure of the repeatability/reproducibility of results rather than their correctness
- *It does not imply anything with respect to their relation to the 'true value'*
- Precision is usually investigated at three levels: repeatability, intermediate precision, and reproducibility

Repeatability

- is a measure of the precision under **the same operating conditions over a short interval of time.**
- It is sometimes referred to as **intra-assay precision**

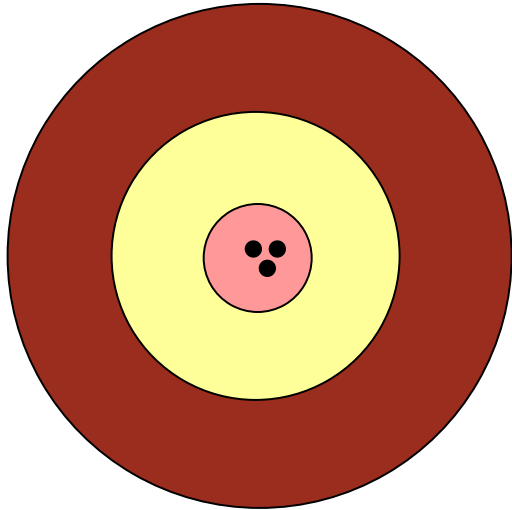
Intermediate Precision.

- Intermediate precision is defined as the variation within the **same laboratory.**
- The extent to which intermediate precision needs to be established depends on the circumstances under which the procedure is intended to be used.
- Typical parameters that are investigated include **day-to-day variation, analyst variation, and equipment variation**

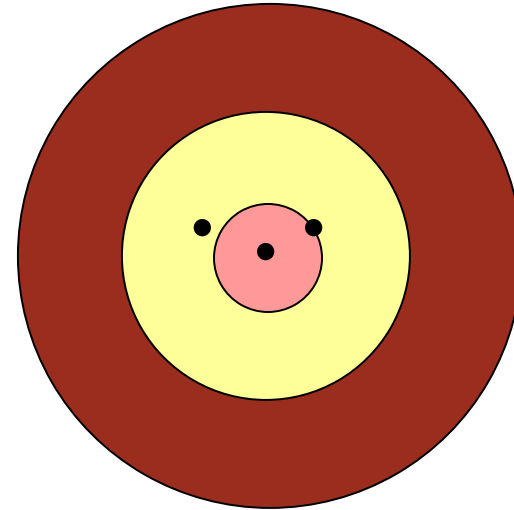
Reproducibility.

- Reproducibility measures the precision **between laboratories**.
- This parameter should be considered in the standardization of an analytical procedure (e.g., inclusion of procedures in pharmacopoeias and method transfer between different laboratories).
- To validate this characteristic, similar studies need to be performed at other laboratories using the same homogeneous sample lot and the same experimental design.
- The most common approach is the ***direct method transfer*** from the originating laboratory to the receiving laboratory.

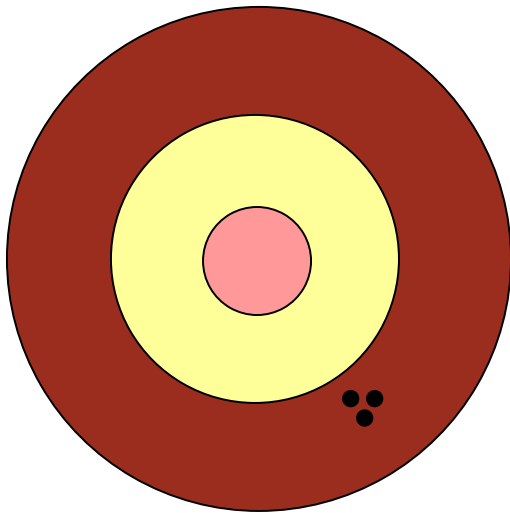
Accuracy & Precision



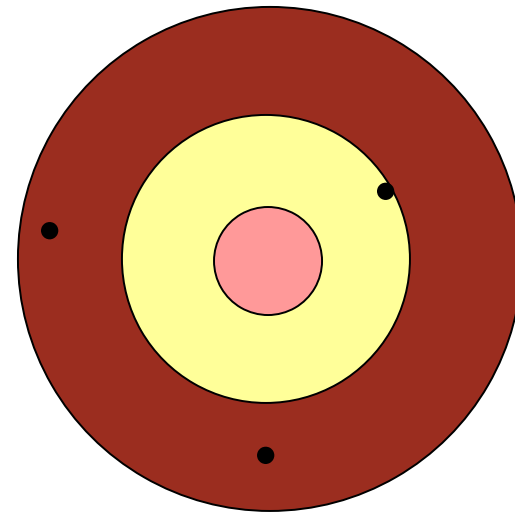
Accuracy without Precision



Precision without Accuracy



No Precision & No Accuracy



Range

- The range of a method is related to its sensitivity.
- The interval between the upper & lower concentration (amounts) of the analyte in the sample (including these concentration) for which it has been shown that the analytical procedure has a suitable level of precision, accuracy & linearity.

Selectivity

- The selectivity of a method is a measure of how capable it is of **measuring the analyte alone** in the presence of other compounds contained in the sample.
- Detection methods can be ranked according to their selectivity.
- A simple comparison is b/n fluorescence & UV spectrophotometry;
 - There are many more compounds w/c exhibit UV absorption than fluorescence, thus fluorescence spectrophotometry is more selective methods.
 - Because selective methods are based on more complex principles than non selective methods, they may be less robust,
 - e.g. fluorescence spectrophotometry is more affected by changes in the analytical method than UV.

Robustness

- The measure of its capacity to remain unaffected by small, but deliberate variations in method parameters and provides an indication of its reliability during normal usage.

Limit of detection

- ICH defines LOD as the lowest amount of analyte in a sample which can be detected but not necessarily quantitated as an exact value.

Limit of quantitation

- ICH defines LOQ of an individual analytical procedure as the lowest amount of analyte in a sample which can be quantitatively determined with suitable precision and accuracy.

1.4. Basic calculations in pharmaceutical analysis

Molarity

- ▶ The molar concentration (C_x) of the solution of the chemical species X is the number of moles of that species that is contained in one liter of the solution (not one liter of the solvent).
- ▶ The unit of molar concentration is molarity, M, which has the dimensions of mol L^{-1} .

$$C_x = \frac{\text{n}^\circ \text{ mole solute}}{\text{n}^\circ \text{ L solution}} = \frac{\text{n}^\circ \text{ m mole solute}}{\text{n}^\circ \text{ ml solution}}$$

- ▶ One liter of one molar solution will consist of one mole of solute plus enough solvent to make a final volume of one liter.

Example:

- ▶ Calculate the molar concentration of ethanol in an aqueous solution that contains 2.30g of $\text{C}_2\text{H}_5\text{OH}$ (46.07 g/mol) in 3.50L of solution.
- ▶ Describe the preparation of 2.00L of 0.108M BaCl_2 from $\text{BaCl}_2 \cdot 2 \text{H}_2\text{O}$ (FW= 244.3g /mol)

Normality

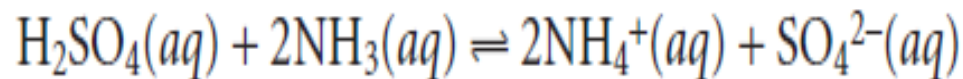
- *Normality* (N) is defined as the number of equivalent weight of solute dissolved in one liter of solution.

$$N = \frac{\text{no_of equivalent weight}}{\text{Volume of so/n in Liter}} \quad \text{no_of equivalent weight} = \frac{\text{weight of solute}}{\text{equivalent weight}}$$

- An equivalent weight is defined as the ratio of a chemical species $EW = \frac{FW}{n}$ la **weight (FW)** to the number of its equivalents
- The number of **equivalents, *n***, is *based on a reaction unit, which is that part of a* chemical species involved in a reaction
- **Normality makes use of the chemical equivalent, which is the amount of one** chemical species reacting stoichiometrically with another chemical species.

1.4. Basic calculations.....

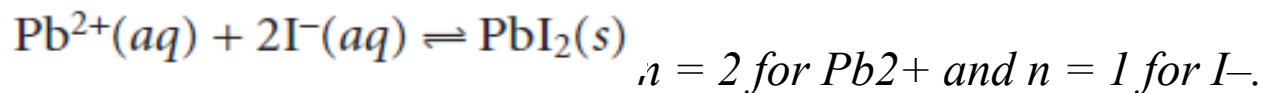
- Note that this definition makes an equivalent, and thus normality, a function of the chemical reaction in which the species participates.
- Although a solution of H₂SO₄ has a fixed molarity, its normality depends on how it reacts.
- In an acid–base reaction, the reaction unit is the number of H⁺ ions donated by an acid or accepted by a base. For the reaction between sulfuric acid and ammonia



n = 2 for H₂SO₄ and n = 1 for NH₃

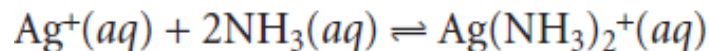
1.4. Basic calculations.....

- In a precipitation reaction, for example, the reaction unit is the charge of the cation or anion involved in the reaction; thus for the reaction



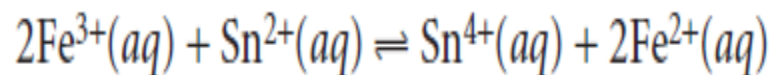
- For a complexation reaction, the reaction unit is the number of electron pairs that can be accepted by the metal or donated by the ligand. In the reaction between Ag⁺ and NH₃

n for Ag⁺ is 2 and that for NH₃ is 1



1.4. Basic calculations.....

- Finally, in an oxidation–reduction reaction the reaction unit is the number of electrons released by the reducing agent or accepted by the oxidizing agent; thus, for the reaction



$n = 1$ for Fe^{3+} and $n = 2$ for Sn^{2+}

Molarity vs Normality

- Molarity: M = moles of solute contained in one liter of solution.
- Normality: N = moles of reactive units per liter (equivalents per liter).
- Normality is always a multiple of molarity.

E.g.

- In an acid-base reaction, normality is a measure of the protons (H^+) or hydroxides (OH^-) that react with one another.
- Consider a 1 M solution of sulfuric acid, H_2SO_4 .
- Normality is a measure of the moles of protons in the solution. Since 2 protons are available to react on each molecule of H_2SO_4 , the normality is 2 N.

Dilutions

- Because solutions in science are often much more concentrated than are desired or can be managed for a given protocol, it is frequently necessary to dilute these solutions to a desired level.
- The number of moles of solute in the concentrated solution is equal to the number of moles of solute in the dilute solution.

$$\text{moles}_{\text{con}} = \text{moles}_{\text{dil}}$$

$$C_{\text{con}} V_{\text{con}} = C_{\text{dil}} V_{\text{dil}}$$

- Dilution factor (DF): ratio of final volume/aliquot volume (final volume = aliquot + diluent)

- **Example:** how would you prepare 0.5 M of HCl acid in 250ml from its stock solution having 11.9 M?

Solution

$$C_{\text{con}} V_{\text{con}} = C_{\text{dil}} V_{\text{dil}}$$

$$11.9\text{M} * V_{\text{con}} = 0.5\text{M} * 250\text{ml}$$

$$V_{\text{con}} = 10.5\text{ml}$$

- By transferring 10.5ml from stock sol to 250ml flask & adding a solvent to the mark.

Weight, Volume, and Weight-to-Volume Ratios

- Weight percent (% w/w), volume percent (% v/v) and weight-to-volume percent (% w/v) express concentration as units of solute per 100 units of sample
- Percent composition of a solution can be expressed as:
 - Weight percent (w/w) = $\frac{\text{mass of solute}}{\text{mass of sol}^n} \times 100\%$
 - Volume percent (v/v) = $\frac{\text{volume of solute}}{\text{volume of solution}} \times 100\%$
 - Weight /volume percent (w/v) = $\frac{\text{mass of solute g}}{\text{volume sol}^n \text{ ml}} \times 100\%$

1.4. Basic calculations.....

- *Weight percent* is frequently employed to express the concentration of commercial aqueous reagents. E.g. 37% hydrochloric solution - this means the reagent contains 37g of HCl per 100g of solution.
- *Volume percent* is commonly used to specify the concentration of a solution prepared by diluting a pure liquid with another liquid.

E.g. 5% aqueous solution of methanol - usually means a solution prepared by diluting 5.0ml of pure methanol to give 100ml of solution with enough water.

- *Weight /volume percent* is after employed to indicate the composition of dilute aqueous solutions of solid reagents.

E.g., 5% aqueous silver nitrate often refers to a solution prepared by dissolving 5g of silver nitrate in sufficient water to give 100ml of solution

Parts per million & parts per billion

- ▶ For very dilute solutions, parts per million (PPM) is convenient way to express concentration:

$$C_{\text{ppm}} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6 \text{ ppm}$$

- ▶ The units of mass in the numerator & denominator must agree.
- ▶ For even more dilute solutions we use parts per billion.

$$C_{\text{ppb}} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^9 \text{ ppb}$$

- ▶ 1ppm is equivalent to 1 mg of solute per liter (mg/l) or 1 mg of solute per kg (mg/kg).
- ▶ E.g. $1\mu\text{g/mL} = 1\text{ppm}$

$$\text{ppm} = \frac{\text{mg}}{\text{liter}} = \frac{\mu\text{g}}{\text{mL}}$$

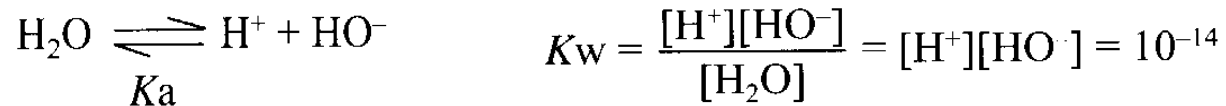
$$\text{ppb} = \frac{\mu\text{g}}{\text{liter}} = \frac{\text{ng}}{\text{mL}}$$

1.5. Physical and chemical properties of drug molecules

- Chemical analysis is an important part of the quality assessment of drugs.
- A deeper understanding of the analytical method requires knowledge about both the analytical technique and the chemical properties of the analytes.
- Therefore a basic knowledge of a number of physicochemical properties of molecules is needed to be able to understand and further develop analytical chemical methods.

Calculation of pH

- The pH of a solution is defined as $-\log [H^+]$, where $[H^+]$ is the conc of hydrogen ion in so/n
- In pure water the conc. of hydrogen ion governed by the equilibrium:



- K_a is the dissociation constant for the equilibrium, is known as K_w in the case of the dissociation of water

- Since the conc of water does not change appreciably as a result of ionization, its conc can be regarded as not having an effect on the equilibrium and it can be omitted from the equation and this means that in pure water:

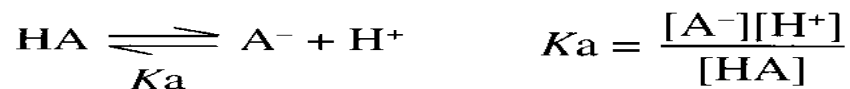
$$[\text{H}^+] = [\text{OH}^-] = 10^{-7}$$

The pH of water is thus given by $-\log 10^{-7} = 7.00$.

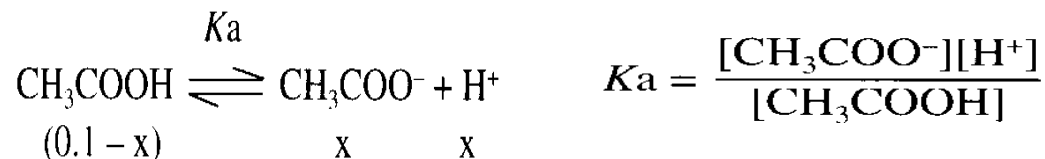
- If an acid is introduced into an aqueous solution the $[\text{H}^+]$ increases

- Strong acid is completely ionized in water and $[H^+]$ is equal to its Molarity
 - e.g. 0.1M HCl contains 0.1M H^+ and has a pH of $\log [0.1] = 1$
- For a so/n of a strong base such as 0.1NaOH, $[OH^-]=1M$ and
 - $[H^+][OH^-] = 1 \times 10^{-14}$, therefore $[H^+] = 1 \times 10^{-13}$ and
 - pH = 13
- Weak acids are not completely ionized in aqueous so/n and are in equilibrium with the undissociated acid, as is the case for water, w/c is a very weak acid.

- The dissociation constant K_a is given by the expression below:



- For instance in a 0.1M sol'n of acetic acid ($K_a = 1.75 \times 10^{-5}$) the equilibrium can be written as follows:



- The pH can be calculated as follows:

$$1.75 \times 10^{-5} = \frac{x^2}{(0.1 - x)}$$

- Since the dissociation of the acetic acid does not greatly change the conc. of the unionized acid, the above expression can be approximated to:

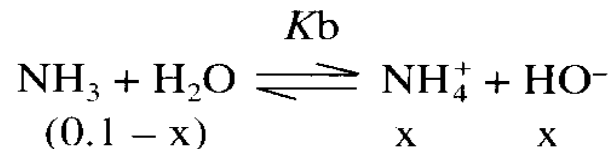
$$1.75 \times 10^{-5} = \frac{x^2}{0.1}$$

$$x = [\text{H}^+] = \sqrt{1.75 \times 10^{-6}} = 0.00132 \text{ M}$$

$$\text{pH} = 2.9$$

- In comparison the pH of 0.1M HCl is 1

The calculation of the pH of a weak base can be considered in the same way. For instance in a 0.1 M solution of ammonia ($K_b = 1.8 \times 10^{-5}$) the equilibrium can be written as follows:



If the concentration/activity of water is regarded as being 1 then the equilibrium constant is given by the following expression:

$$K_b = \frac{[\text{NH}_4^+][\text{HO}^-]}{[\text{NH}_3]}$$

$$1.8 \times 10^{-5} = \frac{x^2}{(0.1 - x)}$$

The concentration of NH_3 can be regarded as being unchanged by a small amount of ionisation and the expression can be written as:

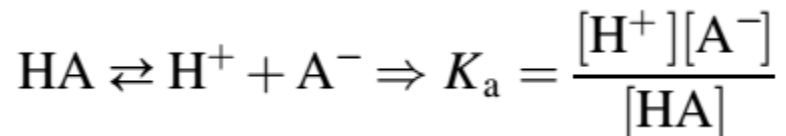
$$x = [\text{HO}^-] = \sqrt{1.8 \times 10^{-6}} = 0.0013 \text{ M}$$

$$[\text{H}^+] = \frac{K_w}{0.0013} = \frac{10^{-14}}{0.0013} = 7.7 \times 10^{-12} \text{ M}$$

$$\text{pH} = 11.1$$

In comparison the pH of 0.1 M NaOH is 13.

pKa

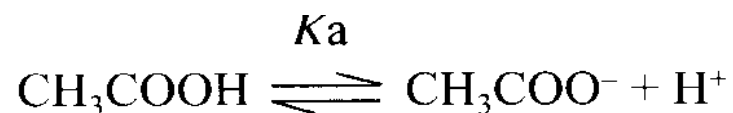


- The pKa value of a cpd is defined as:

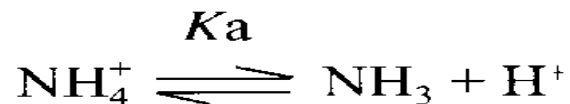
$$\text{pK}_a = -\log K_a$$

- If pKa is used as a measure of acidic or basic strength,
 - For an acid the smaller the pKa value the strongest the acid.
 - For a base the largest the pKa value the stronger the base.

- For an acid the forward reaction used



- In the case of a base it is the protonated form of the base that act as a proton donor



Buffer solution

- A solution containing a weak acid/ base and its conjugate base/acid that is resistant to a change in pH when a strong acid or strong base is added.
- Adding as little as 0.1 mL of conc HCl to a liter of H₂O shifts the pH from 7.0 to 3.0 ($\Delta\text{pH} = 4$).
- The same addition of HCl to a liter solution that is 0.1 M in both a weak acid and its conjugate base, however, results in only a negligible change in pH ($\Delta\text{pH} = 0.09$).

- ▶ A mixture of acetic acid and sodium acetate is one example of an acid/base buffer
- ▶ The equilibrium position of the buffer is governed by the reaction



$$K_a = \frac{[\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = 1.75 \times 10^{-5}$$

- ▶ The relationship between the pH of an acid-base buffer and the relative amounts of CH_3COOH and CH_3COO^- is derived by taking the negative log of both sides of the above equation and solving for the pH

$$\text{pH} = \text{p}K_a + \log \frac{[\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = 4.76 + \log \frac{[\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$$

- ▶ Buffering occurs because of the logarithmic relationship between pH and the ratio of the conjugate base and weak acid concentrations.
- ▶ For example, if the equilibrium concentrations of CH_3COOH and CH_3COO^- are equal, the pH of the buffer is 4.76.
- ▶ If sufficient strong acid is added such that 10% of the acetate ion is converted to acetic acid, the concentration ratio $[\text{CH}_3\text{COO}^-]/[\text{CH}_3\text{COOH}]$ changes to 0.818, and the pH decreases to 4.67.

- A more useful relationship relates the buffer's pH to the initial concentrations of weak acid and base.
- A general buffer equation can be derived by considering the following reactions for a weak acid, HA, and the salt of its conjugate base, NaA.

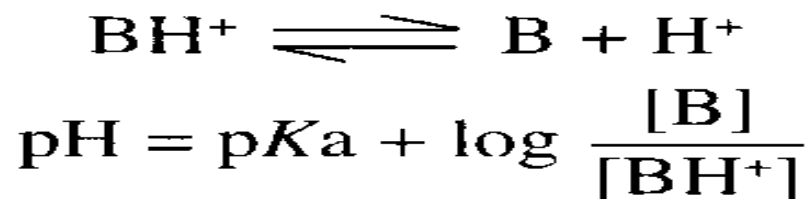


- After several rearrangement, it provide us a general formula known as henderson-Hasselbalch equation

$$\text{pH} = \text{p}K_a + \log \frac{C_{\text{NaA}}}{C_{\text{HA}}}$$

- Where C_{NaA} conc of salt, C_{HA} conc of the acid

- Hasselbalch equation provides a simple way to calculate the pH of a buffer and to determine the change in pH upon adding a strong acid or strong base.
- For a base, Henderson-Hasselbalch equation is written as



- Using Hasselbach equation it is possible to determine degree of ionization of a drug at a given pH.
- E.g. degree of ionization of acetic acid at pH of 4.76

$$\text{pH} = \text{pK}_a + \log \frac{C_{\text{NaA}}}{C_{\text{HA}}} \quad \text{pH} = 4.76 + \log \frac{[\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$$

$$4.76 = 4.76 + \log \frac{[\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} \quad \log \frac{[\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = 0$$

$$\frac{[\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = 10^0 = 1 \quad \text{acetic acid ionized 50\% at pH of 4.76}$$

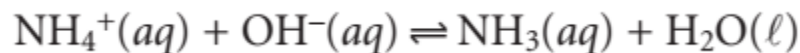
- E.g1. Calculate the pH of a buffer that is 0.020 M in NH_3 and 0.030 M in NH_4Cl . What is the pH after adding 1.00 mL of 0.10 M NaOH to 0.10 L of this buffer? ($K_a = 5.7 \times 10^{-10}$)

Solution

- Initial pH of the buffer is:

$$\text{pH} = 9.24 + \log \frac{C_{\text{NH}_3}}{C_{\text{NH}_4^+}} = 9.24 + \log \frac{0.020}{0.030} = 9.06$$

- Adding NaOH converts a portion of the NH_4^+ to NH_3 due to the ff reaction:



- The new concentrations of NH_4^+ & NH_3 are:

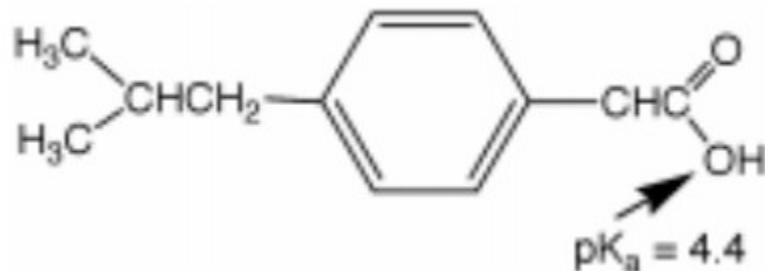
$$\begin{aligned} C_{\text{NH}_4^+} &= \frac{\text{moles NH}_4^+ - \text{moles OH}^-}{V_{\text{tot}}} \\ &= \frac{(0.030 \text{ M})(0.10 \text{ L}) - (0.10 \text{ M})(1.00 \times 10^{-3} \text{ L})}{0.101 \text{ L}} = 0.029 \text{ M} \end{aligned}$$

$$\begin{aligned} C_{\text{NH}_3} &= \frac{\text{moles NH}_3 + \text{moles OH}^-}{V_{\text{tot}}} \\ &= \frac{(0.020 \text{ M})(0.10 \text{ L}) + (0.10 \text{ M})(1.00 \times 10^{-3} \text{ L})}{0.101 \text{ L}} = 0.021 \text{ M} \end{aligned}$$

- Final pH of the buffer is:

$$\text{pH} = 9.24 + \log \frac{0.021}{0.029} = 9.10$$

- **E.g2** How many moles of sodium acetate and acetic acid are required to prepare 1 liter of a buffer, pH 5.0, which is 0.1 M in total available acetate (dissociated and undissociated). Acetic acid has a pK of 4.74.
- **Solution:** From Buffer Equation ,
$$5.0 = 4.74 + \log [\text{salt}] / [\text{acid}]$$
or $\log [\text{salt}] / [\text{acid}] = 0.26$.
$$[\text{salt}] / [\text{acid}] = 1.82 , [\text{salt}] + [\text{acid}] = 0.1$$
- Acid = $0.1 / 2.82 = 0.035\text{M}$
- Salt = $0.1 \times 1.82 / 2.82 \text{ mole} = 0.065 \text{ mole}$.



- Have you ever taken an over-the-counter (OTC) drug called ibuprofen?
 - with the carboxylic acid moiety (COOH) the proton donor, which has a pK_a of around 4.4.
 - What happens once it's swallowed?

Solution

$$\% \text{ ionization of acidic drug} = 100 \times \frac{[\text{salt}]}{[\text{acid}] + [\text{salt}]}$$

- First it goes to the stomach. We'll assume the pH of the stomach is around 3.

$$\text{pH} = \text{pK}_a + \log \frac{[\text{salt}]}{[\text{acid}]}$$

$$3 = 4.4 + \log \frac{[S]}{[A]}, \therefore \frac{[S]}{[A]} = 10^{-1.4} = 0.03981$$

$$\% \text{ Salt} = \frac{0.03981}{1.03981} \times 100 = 4\%$$

- Most of the drug is in acid form and therefore could cross the stomach lining to enter the bloodstream; i.e., ibuprofen can be absorbed from the stomach.

- Next the large intestine, assuming pH = 8

$$8 = 4.4 + \log \frac{[S]}{[A]}, \therefore \frac{[S]}{[A]} = 10^{3.6} = 3981$$

$$\% \text{ Salt} = \frac{3981}{3982} \times 100 = 99.97\%$$

- Most of the drug is in salt form and so shouldn't readily cross the intestinal lining and enter the bloodstream.
 - This is not true in real life, however. About 80% of ibuprofen is absorbed. Why so much? Partly because there is a very large surface area in the gut, maximizing the amount of unionized drug coming into contact with the membrane.
 - Partly because some absorption takes place in the duodenum, where the pH is lower (closer to 6) and more drug is unionized. There may be some that crosses the membrane via pores, since it is not a very large molecule (MW ≈ 187). The most likely reason that the drug is pretty well absorbed, though is by active transport across the gut wall.

Stability of drugs

- Many drugs are quite stable but functional groups such as esters and lactam rings w/c occur in some drugs are susceptible to hydrolysis and functional groups such as catechols and phenols are quite readily oxidized.
- The most common type of degradation w/c occur and formulated drugs obey zero or first order kinetics

Zero order degradation

- ▶ In zero order kinetics the rate of degradation is independent of the conc of the reactant.
- ▶ This type of degradation is typically of hydrolysis of drugs in suspension or tablets where the drugs is initially in the solid state and gradually dissolve at the same rate as the drug in so/n is degraded

First order degradation

- This type of degradation would be typical of hydrolysis of a drug in so/n
- In first order kinetics the rate constant k has units h^{-1} or s^{-1} and the rate of the reaction for a drug is governed by the expression

$$-\frac{d[A]}{dt} = k[A]$$

- From this expression by integration and rearrangement the following expression arises:

$$t = \frac{1}{k} \ln \frac{a}{a-x}$$

- The half life of the drug (the time taken for 50% of a sample drug to degrade, i.e. where x is $a/2$) is thus given by the following expression

$$t_{0.5} = \frac{1}{k} \ln \frac{a}{a/2} = \frac{1}{k} \ln 2$$

Thus for aspirin, which has a rate constant of 0.0133 h^{-1} for the hydrolysis of its ester group at 25° and pH 7.0, the half-life can be calculated as follows:

$$t_{0.5} = \frac{0.693}{0.0133} = 52.1 \text{ h}$$

The shelf-life of a drug, the time required for 10% degradation (where x is 0.1 a), is given by the following expression:

$$t_{0.9} = \frac{1}{k} \ln \frac{a}{0.9a} = \frac{1}{k} \ln 1.11$$

$$t_{0.9} = 7.85 \text{ h}$$